



Mini Project – Set up NodeJS environment

Using the resources and lessons from this week, you are being tasked with setting a development environment for a NodeJS app. You are not expected to know NodeJS and may need to do some researching into it.

You have been asked to setup Terraform/Ansible configurations for the following:

- VPC with 2x public subnets, each with 400+ available IP addresses
- MySQL RDS with configured credentials
- 1x CI Server EC2 with ports open for HTTP, MySQL, Jenkins and SSH
- 1x Deployment EC2 with public HTTP and MySQL, and SSH (from the control plane where you're running the setup AND from the CI Server)
- CI Server should have Docker, Jenkins, unzip, wget and curl installed
- CI Server needs to add Jenkins to sudo group and add user to Docker group
- Deployment EC2 should run the app at this repo <https://gitlab.com/Reece-Elder/devops-m5-nodeproject/-/tree/main>

Note: You do not need to actually deploy any resources – the goal of this activity is simply to set up the configs.

You will be tasked within groups to do this, and you should be using a version control system to push your code up to. Within your VCS you should take advantage of proper Feature Branch Management and merge requests.

Each group will be asked to give a brief (5-10 minute) presentation overviewing their solution. This presentation should cover:

- Architecture: what will the deployment look like, architecturally speaking?
- Modularisation: What Ansible roles/terraform modules have you created as part of this solution?
- Integration: How will you orchestrate the integration of terraform and ansible via the CI Server

Stretch goal – How might you adapt your solution to use Docker to containerise and deploy the NodeJS app?